IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 10: Gradient Magnitude Calculation (Edge Detection)

Given Image Patch (3×3):

[10 10 10]

[0 0 0]

[0 0 0]

Given Sobel Operators Gx and Gy (as discussed in class).

Tasks:

1. Compute Gx.

2. Compute Gy.

3. Calculate Gradient Magnitude: G = sqrt(Gx^2 + Gy^2).

4. Identify whether it is a horizontal or vertical edge.

Answer:

1. Gx = 0

2. Gy = -40

3. Gradient Magnitude: $G = \sqrt{(Gx^2 + Gy^2)} = \sqrt{(0^2 + (-40)^2)} = \sqrt{1600} = 40$

4. It is a horizontal edge because the intensity changes mainly in the vertical direction (bright pixels at the top and dark pixels below).